Lucas Brown

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Control Software Engineer with five years' experience spanning control theory, simulation and real-time implementation across fusion, robotics and automotive.

Skills

Engineering	Control Theory, Software Development, Plasma Physics, System Identification/Machine Learning, Communication Protocols, Embedded Systems Engineering.
Technical Tools	Torax, FreeGS, Python (Jax, PyTorch), C++, Julia, Mathematica, MATLAB, Linux/UNIX (Bash, CLI), Git, HDL, ROS2, NI DAQmx C/C# API, Simulink, C#, Mbed C, COMSOL.
Communication	B2 French, A2 Chinese, Requirements Capture, Presenting and Work Package Management.

Experience

UK Atomic Energy Authority - Control Systems Software Engineer 2021 - present

- Control Theory research and simulation:
  - Identified non-linear control orientated models for fluid dynamics prediction with Bayesian and deterministic machine learning. Using these models, I designed and simulated a controller in MATLAB/Simulink. Presenting the results at Symposium on Fusion Technology 2026.
  - Simulated multi-robot payload handling scenario in Julia, controlled with Model Predictive Control.
  - Designed and implemented controllers for non linear fluid and robotic systems using MATLAB/Python.
- Control Systems application:
  - Programmed a gas flow controller onto a Schneider Electric PLC. Implemented a gas plant simulator on a FPGA-CPU board for hardware-in-the-loop testing, integrated with Modbus TCP, synchronised via NTP.
  - Developed and validated a configurable C++ ROS2 node to acquire real-time samples from National Instruments hardware. Managed Linux real-time kernel modules for integration with NI packages.
  - Implemented an Optimal Control scheme onto a distributed robotic system, integrating sensors and Beckhoff embedded PCs through CANOpen over EtherCAT and ADS over TCP/IP. Integrated with Python ROS2.
  - Led a team to develop a PyQt HMI application for a robotic bench test, to start and record experiments.
- Fusion Experience:
  - Studied plasma control, transport codes and equilibrium states at EPFL Tokamak Control course.
  - Managed a multi-disciplinary team to integrate sensors into MAST-U's data acquisition system during campaign.
  - Currently driving upgrade of MAST-U gas introduction modules for exhaust capabilities in MU06.
- Publications: *Baker, H., Brown, L., Parrott, A., On the use of LSTM-based estimation components for tokamak gas actuator control, Fusion Engineering and Design, Volume 215, 2025.* Three more papers submitted.

CEA Cadarache - Control Systems Engineering Secondee 2024-2025

- Collaborated with international partners to develop modern software for fusion inspection robots. Jointly publishing progress and results at Symposium on Fusion Technology 2026.
- Developed a containerised, modular and interoperable C++ robotic simulation with factory design pattern.
- Built a robust environment with layered Docker, rostest and gtest for Continuous Integration.

Mercedes AMG Petronas Formula 1 - Electronic Development Industrial Placement 2020

- Configured an automated calibration bench test tool for pre-race validation of sensors using SCPI.

Rolls-Royce Submarines - Electrical Control & Instrumentation Intern 2019

- Interfaced with stakeholders to define specifications and to design the architecture of a digital radiation sensor.

Education

Imperial College London 2017-2021

- Awarded Maurice Hancock Scholarship & First Class Honours Master's in Electrical & Electronic Engineering.
- Key Modules: Predictive Control, Digital Control Systems, Digital Signal Processing & Power Systems.
- Key Projects: Applied model predictive control on gantry crane system for robust payload manipulation.
- Developed a high dynamic range signal emulation system on FPGA for biomedical hardware-in-the-loop testing.

Personal Projects

- Developed a generic Rubik's Cube simulation in Python with Tkinter GUI. Developing Group Theory and Reinforcement Learning (Q-Learning) solvers, targeting deployment on Zynq FPGA hardware.